INTERNATIONAL MONETARY FUND

DEMOCRATIC REPUBLIC OF TIMOR-LESTE

March 8, 2019

STAFF REPORT FOR THE 2019 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS[1]

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Prepared by Staff of the International Monetary Fund and the International Development Association

Risk of external debt distress	Low
Overall risk of debt distress	Low
Granularity in the risk rating	Substantial space to absorb shocks
Application of judgment	Yes, availability of large and liquid financial assets mitigate debt/debt service needs

Timor-Leste's risk of external debt distress is assessed as low, using judgment that takes account of the country's financial assets in the Petroleum Fund (PF). The assessed risk is improved from the previous DSA's finding of "moderate risk", reflecting fuller considerations of risk mitigating elements. Timor-Leste's Public and Publicly Guaranteed (PPG) external (and public) debt is small (5 percent of GDP in 2018), and debt indicators are projected to remain below their thresholds under the baseline scenario over the entire projection period through 2038. Standardized stress tests show that a shock to the primary balance is the most extreme shock to the debt trajectory, causing a short-lived and marginal breach of the threshold for the PV of PPG debt-to-exports ratio. However, the Petroleum Fund (PF)—estimated at 506 percent of GDP in 2018— is large relative to projected debt levels and debt service requirements; and its assets are liquid and accessible, prompting the use of judgment to upgrade the risk assessment. Beyond the 20-year projection horizon, debt is projected to rise quickly, creating risks to debt sustainability. The authorities should adopt prudent policies to address this risk. The DSA illustrates the benefits of such policies in an alternative scenario.

[1] This Debt Sustainability Analysis DSA was prepared jointly with the World Bank, in accordance with the revised Debt Sustainability Framework for low-income countries approved by the Executive Boards of the IMF and the IDA. Timor-Leste's Composite Indicator (CI) index indicates that the country's debt-carrying capacity is medium. Thresholds for external debt-burden indicators are set at levels compatible with the new framework: PV of debt-to-GDP ratio (40), PV of debt-to-exports ratio (180), debt service-to-exports ratio (15) and debt service-to-revenue ratio (18). According to the World Bank's operational policies, Timor-Leste is classified as a 'Blend' country, which means that it is IDA-eligible, but also creditworthy for some IBRD borrowing.

PUBLIC DEBT COVERAGE

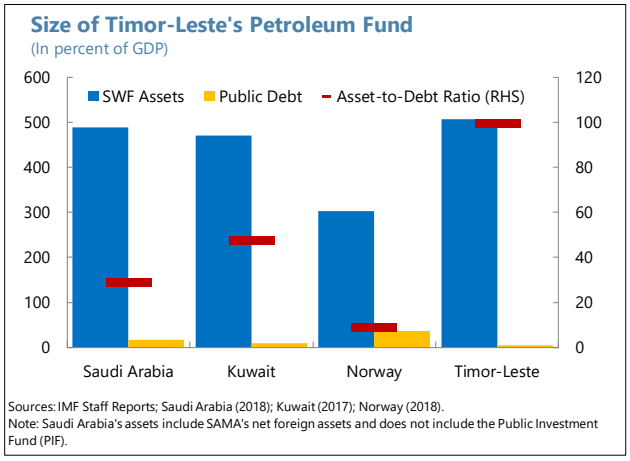
1. The coverage of public sector debt used in this report is public and publicly guaranteed (PPG) debt. Timor-Leste's PPG external debt is held entirely by the central government. The coverage of public sector debt includes SOE debt. Under the Public Debt Regime Law, SOEs are not allowed to borrow for themselves and can only obtain financing via on-lending from the Ministry of Finance.[2] The public sector only borrows externally, given a lack of domestic financing sources. The Timorese private sector has negligible medium- and long-term external liabilities. The debt definition of the DSA is currency-based and the legal tender is the US dollar.

	Subsectors of the public sector	Check box
1	Central government	X
2	State and local government	
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	
8	Non-guaranteed SOE debt	

BACKGROUND ON DEBT

2. Timor-Leste's net public asset position is currently strong due to oil-related savings accumulated in the Petroleum Fund (PF).

- *Size of the PF*. The PF balance at end-2018 stood at US$15.8 billion (506 percent of total GDP and 848 percent of total non-oil GDP), covering more than 166 months of goods and services imports. This makes the PF one of the largest Sovereign Wealth Funds (as percent of GDP) in the world (see figure). The PF closing balance of 2017 improved compared to 2016 on the back of higher inflow of oil and gas receipts, favorable investment returns and valuation effects, and lower government spending due to a political impasse. However, heightened global financial market volatility in late-2018 led to negative valuation effects and a decline in the closing balance at end-2018.

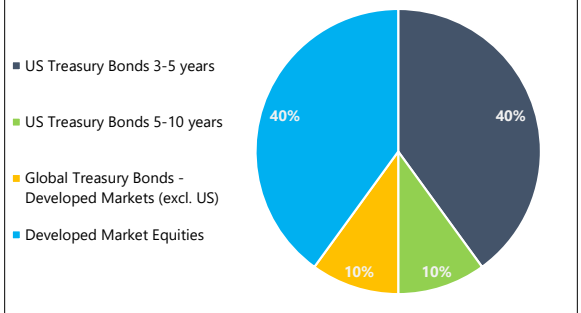


Sources: IMF Staff Reports; Saudi Arabia (2018); Kuwait (2017); Norway (2018).
Note: Saudi Arabia's assets include SAMA's net foreign assets and does not include the Public Investment Fund (PIF).

[2] According to the Public Debt Regime Law No. 13/2011, the Government of Timor-Leste (GoTL), in particular, the Ministry of Finance, is the only entity that may engage in borrowing, motivated by financing needs generated by the need to execute the State's priority tasks relating to the building of strategic infrastructures for the country's development.

- *Liquidity*. The Petroleum Fund Law sets out eligible investments, guidelines for asset allocation, and risk limits for the PF. Given the annual withdrawals from the PF to the budget, the PF's targeted asset allocation reflects a preference for highly liquid assets (see figure). Currently, the targeted asset allocation is 60 percent in Treasury Bonds of developed countries and 40 percent in developed market equities.[3] Since its inception, the average return on PF assets has been around 4.4 percent (as of end-2017).[4]

Timor-Leste: Petroleum Targeted Asset Allocation, 2017
(In percent)



Source: Timor-Leste Petroleum Fund Annual Report (2017).

- *Accessibility*. The PF constitutes the main financing source for the budget. The amount is guided by the Estimated Sustainable Income (ESI), which is set at 3 percent of total Petroleum Wealth[5]. Withdrawals in excess of the ESI can be made, provided that the Parliament approves the government's explanation that this is made in the long-term interests of the country. Over the lifetime of the PF, the government has withdrawn a total of $9.6 billion (end-2017), with excess withdrawals accounting for about 63 percent of this amount. The parliament has yet to reject a request for excess withdrawals.[6]

3. Outstanding public external debt is projected at US$157.7 million (5.1 percent of total GDP and 8.5 percent of total non-oil GDP) at end-2018. External loans signed totaled US$355 million (12.8 percent of total GDP and 19 percent of total non-oil GDP) by end-2018, which constitutes twelve loans under eight packages and will be disbursed over the 4 years until 2022. Timor-Leste's external borrowings are from the Asian Development Bank, Japan International Cooperation Agency (JICA) and the World Bank Group. These loan packages are to support infrastructure projects, primarily assigned for rehabilitation and upgrading of national roads (Text

Text Table 1. Timor-Leste: External Public Debt Committed by Creditor Groups
(In millions of U.S. dollars)

	2016	2017	2018
Multilateral	271	355	355
Bilateral Non-Paris Club	50		
Total	321	355	355

Source: Timor-Leste Ministry of Finance.

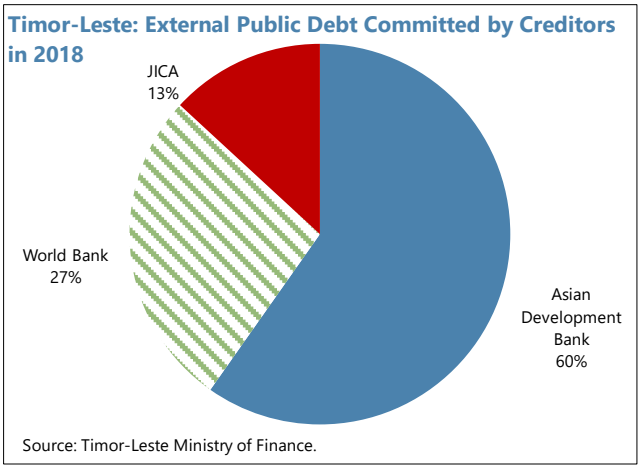
[3] The 2011 Petroleum Fund Law specifies that no more than 5% of the Petroleum Fund should be invested in other eligible investments. The Council of Ministers recently approved a resolution to use these 5 percent to enable the purchase of the stakes of ConocoPhillips and Shell in the Greater Sunrise Joint Venture ($650m).

[4] Staff assumes a 3.9 percent return on assets when projecting the PF balance.

[5] Petroleum Wealth comprises the balance of the Petroleum Fund and the Net Present Value of expected future petroleum revenue

[6] According to the constitution, the President has the right to veto a budget that has been approved by the parliament. However, within 90 days, the Parliament can overrule the veto by an absolute majority of its members in full exercise of their functions (see Section 88, paragraph 2 of the Constitution of The Democratic Republic of Timor-Leste).

Table 1).[7] The Asian Development Bank has the largest share of total external debt, comprising nearly two-thirds of total external debt of Timor-Leste in 2018 (Chart below).



BACKGROUND ON MACRO FORECASTS

4. This DSA is based on the macroeconomic projections underlying the 2019 Article IV consultation. To illustrate the impact of different policy options on debt sustainability, two scenarios—baseline and reform—are considered.

5. Compared to the previous DSA,[8] there have been some changes (in some cases, substantial) in assumptions underlying the current DSA, due to factors that altered staff's macroeconomic and fiscal assumptions. First, political impasse beginning mid-2017 prevented a supplementary budget for 2017 from passing and resulted in a delay of the 2018 budget approval process. As a result, capital spending envisaged for 2017 under the 2017 budget was not fully executed, leading to a contraction of real GDP in 2017. Second, under the 2018 and 2019 budgets, the medium-term trajectory of government spending is substantially lower than was laid out in the 2017 budget. In particular, a front-loading of capital expenditure with ambitious infrastructure project plans in the 2017 budget was significantly reduced. These factors explain deviations of the baseline assumptions on macroeconomic and fiscal assumptions from the previous DSA exercise (Table 1).

- **Real non-oil GDP** for 2017 is expected to have contracted significantly, due to lower government capital spending and failure to pass a supplementary budget. Political uncertainty faded in mid-2018, and growth is expected to rebound in 2018 to 0.8 percent. Over the long-term, growth is projected to stabilize at around about 5 percent, only marginally lower than in the previous DSA. Oil production from active fields is projected to cease in 2022.

[7] In December 2015, the loan agreement signed by the Government of Timor-Leste and China EXIM Bank did not obtain the approval of the Audit Court due to legality issues on the project for the upgrade of the drainage infrastructure in Dili.

[8] See Timor-Leste 2017 Article IV Staff Report, IMF Country Report No. 17/360.

- **Inflation** is expected to increase steadily to about 4 percent over the medium-term on account of higher global food and fuel prices and spillovers from public investment activity.
- **The current account balance** is projected to remain in deficit over the medium term. Declining income from oil and gas receipts contributes to the deficit. Overall, the current account balance deficit between 2018-2023 is projected to be less severe compared to the previous DSA. This is due to a lower public expenditure trajectory under the 2019 budget, which translates into lower demand for imports and an improved trade balance.
- **The primary fiscal balance** is projected to remain in deficit of about 13 percent of GDP in 2018. The primary deficit is projected to narrow gradually to about 11 percent on average over the long-term (2028-2038). Capital spending is projected to grow to fill the significant infrastructure gap.
- **External financing** consists of concessional loans from official creditors. Private external borrowing is assumed to remain negligible. Timor-Leste does not have exceptional financing such as accumulation of arrears.
- **Fiscal financing** consists of (i) PF withdrawals in excess of ESI and (ii) external borrowing.
- **The grant element of loans** is assumed to decline moderately over the medium-term as the economy develops.
- **Other assumptions** include that no off-budget debt is accumulated including by state-owned enterprises in line with existing legislation.

Table 1. Timor-Leste: Key Macroeconomic and Fiscal Assumptions

	Current (2019 Article IV)				Previous (2017 Article IV)			
	2017	2018	2018-23	2024-38	2016	2017	2017-22	2023-37
Real GDP growth (in percent)	-4.5	-8.0	-5.6	4.8	-7.9	-8.0	-3.7	5.2
Real non-oil GDP growth (in percent)	-4.6	0.8	4.2	4.8	5.5	3.0	5.0	5.2
Inflation (CPI annual average, percent)	0.5	2.3	3.2	4.0	-1.3	1.0	3.2	4.0
Revenues (excluding grants, percent of GDP)	30.7	29.7	26.3	19.1	27.5	24.2	22.3	15.9
Current expenditure (percent of GDP)	33.5	28.2	35.8	24.6	36.3	33.4	35.1	23.1
Net acquisition of non-financial assets (percent	9.5	12.8	10.4	4.7	22.1	10.9	20.2	5.4
Net lending/borrowing (percent of GDP)	-18.7	-17.2	-24.3	-14.1	-30.8	-20.1	-33.3	-12.9
Net incurrence of liabilities (percent of GDP)	1.1	2.0	2.3	2.6	1.1	0.4	5.0	2.1
Exports of G&S (y/y growth)	12.6	8.4	12.5	17.5	5.7	-1.6	14.6	20.8
Imports of G&S (y/y growth)	-11.9	7.0	2.9	7.3	-11.1	-0.2	6.1	6.1
Current account balance (percent of GDP)	-10.2	-8.9	-4.1	-6.5	-18.9	-3.4	-15.9	-4.1

Box 1. Macroeconomic Assumptions Underlying the DSA Update

The macroeconomic assumptions underpinning the current DSA incorporate the economic impacts of the political stalemate beginning 2017, which led to a failure to pass a rectification budget for 2017 and a delay in the approval of the 2018 budget. As a result, the macroeconomic assumptions used for the current DSA exhibit significant deviations from those used for the previous DSA.

Macro Indicators	Last Historic Year (2017)	First Forecast Year (2018)	Medium- and Long-Term Averages
GDP growth	Real non-oil GDP growth for 2017 is estimated to be -4.6 percent. Public spending experienced an abrupt halt in the second half of the year, as the minority government failed to pass a rectification budget.	Growth is expected to recover mildly in 2018, as the political situation improved after the snap election in May. However, with a moderate spending envisioned for 2018 under the budget, the recovery is expected to be weak, at around 0.8 percent.	Over the long-term, growth is expected to stabilize at around 5 percent, slightly lower than in the previous DSA.
CPI Inflation (annual average, percent)	CPI inflation in 2017 was generally weak at 0.5 percent.	CPI inflation is expected to reach 2.3 percent in 2018, due to higher import prices.	CPI inflation is projected to stabilize at around 4 percent over the medium-term.
Current Account	The estimated current account balance for 2017 is revised downward from -3.4 percent in the 2017 DSA to -10.2 percent.	The current account deficit is expected to narrow in 2018.	The current account balance is expected to remain in deficit over the medium term with lower oil and gas receipts and imports of goods and services generated by demand from capital expenditure.
Fiscal Position	The primary deficit in 2017 is 7 percent of GDP, (33 percent in previous DSA). This reflects a much lower level of actual public spending executed for 2017 than the spending scheduler under the 2017 budget.	The primary deficit in 2018 is projected to be 13 percent of GDP.	The primary deficit is projected to narrow gradually to about 3 percent in the long-term.

6. The new realism tools suggest that macroeconomic and fiscal assumptions are reasonable (Figure 4). The three-year adjustment in the primary balance is expected to be a loosening of 5 percent of GDP, between 2017 and 2020. This represents significant stimulus to the economy, and lies near the bottom of the distribution of the realism tool. With regards to the realism tool on fiscal adjustment and growth, a high fiscal multiplier for 2019 partly shows the low base effect of growth in 2018. Lastly, we do

not have relevant data (stocks of government capital) to test other sets of realism tools, and efforts will be made to collect the data for the next DSA.

7. The reform scenario illustrates staff's proposed strategy to safeguard fiscal sustainability. The scenario is broadly similar to staff's recommendations in the 2016 and 2017 Article IV consultation Staff Reports (see section A in the Staff Report for more details). The reform scenario includes:

- *Less front-loading of public expenditure.* The less ambitious front-loading of capital investment as laid out in the 2019 budget is maintained.
- *Domestic revenue.* A value-added tax (VAT) is implemented by 2022. The VAT is estimated to generate revenue of about 3¾ percent of GDP based on a VAT rate of 10 percent in the medium term. VAT revenue increases to at least 5 percent of GDP in the long term.
- *Concessional borrowing and PF excess withdrawals***.** PF withdrawals in excess of the ESI are not be permitted from 2028 and onwards. The resulting financing gap is met by external borrowing.
- Under this reform scenario, higher domestic revenue coupled with no excess withdrawal from 2028 reverse the decline in PF assets, with the PF balance increasing to close to US$13.5 billion (385 percent of total GDP) in 2023, and reaching around US$12.8 billion (75 percent of total GDP) in 2038.
- Overall, external borrowing under the reform scenario is higher than in the baseline in the long-term. In the medium-term (until 2028 when excess withdrawals are permitted), the financing gap under the reform scenario is smaller than the baseline scenario starting 2022, due to the revenue collected from VAT. As a result, the average excess withdrawals over the medium-term (2020-2027) are reduced from US$740.7 million (baseline) to US$618.3 million (reform scenario). From 2028, external borrowing under the reform scenario increases despite domestic revenue from VAT, as the excess withdrawals from the Petroleum Fund are no longer allowed. As a result, the average borrowing over the long-term (2028-2038) rises to US$805.6 million, higher than the projected borrowing of US$279 million under the baseline scenario (Table 3).

Table 2. Timor-Leste: Macroeconomic Assumptions: Baseline and Reform Scenarios

	Baseline Scenario				Reform Scenario			
	2017	2018	2018-23	2024-38	2017	2018	2018-23	2024-38
Revenues (percent of GDP)	30.7	29.7	25.9	16.6	30.7	29.7	27.1	22.5
Current expenditure (percent of GDP)	33.6	28.2	34.1	25.9	33.5	28.2	34.1	25.9
Net acquisition of non-financial assets (percent of GDP)	-17.7	-15.3	-20.6	-11.6	-17.7	-12.4	-25.7	-3.3
Net lending/borrowing (percent of GDP)	-18.7	-17.2	-22.9	-14.2	-18.7	-17.2	-21.6	-10.0
Net incurrence of liabilities (percent of GDP)	2.5	2.0	2.3	2.6	1.1	2.0	2.3	6.7
Borrowing (in millions of USD, period average)	70.0	61.6	75.5	237.2	30.1	61.6	52.0	516.5
Exports of G&S (y/y growth)	12.6	8.4	12.5	17.5	12.6	8.4	12.5	17.5
Imports of G&S (y/y growth)	-11.9	7.0	2.9	10.2	-11.9	7.0	2.9	10.2
Current account balance (percent of GDP)	-10.2	-11.2	-5.3	-15.2	-10.2	-11.2	-5.2	-13.0
External Debt (percent of GDP)	3.8	5.1	10.4	21.4	3.8	5.1	10.4	41.3
Petroleum Fund Balance (in millions of USD)	16,799	15,803	14,943	8,116	16,799	15,803	15,004	13,187

Table 3. Timor-Leste: Projected Medium-Term Fiscal Funding Gaps: Baseline and Reform Scenarios
(In millions of US Dollars)

	2018	2019	2020	2020-2027	2028-2038
Baseline Scenario					
Funding Gap	493.7	1344.1	982.7	839.9	1199.6
Excess Petroleum Fund withdrawal	432.1	1257.1	935.1	740.7	920.6
Borrowing	61.6	87.0	47.5	99.2	279.0
Reform Scenario					
Funding Gap	493.7	1344.1	982.7	715.0	805.6
Excess Petroleum Fund withdrawal	432.1	1257.1	935.1	618.3	0.0
Borrowing	61.6	87.0	47.5	96.7	805.6
Memorandum item					
2019 Budget Framework					
Funding Gap	493.7	1099.4	1451.0	n.a	n.a
Excess Petroleum Fund withdrawal	432.1	1012.4	1411.0	n.a	n.a
Borrowing	61.6	87.0	40.0	n.a	n.a

COUNTRY CLASSIFICATION AND DETERMINATION OF SCENARIO STRESS TESTS

8. The country's debt-carrying capacity applied in the 2019 DSA is medium. The Composite Indicator (CI) index, which has been calculated based on the October 2018 WEO and the 2017 Country Policy and Institutional Assessment (CPIA), is 2.72, indicating that the country's debt-carrying capacity is medium. Prior to this update, the thresholds for previous DSAs were based on the CPIA index, where Timor-Leste was classified as "weak" in terms of policy performance.

9. Given the heavy reliance of oil and gas exports in Timor-Leste's total exports, commodity price shocks are introduced in a tailored stress test. Lack of export diversification and weak external competitiveness results in excessive reliance of Timor-Leste's external position on petroleum and gas receipts. In 2018, oil and gas exports comprised of 77 percent of total exports of goods and services. No adjustments have been made to the default settings regarding the scenario stress testing.

Calculation of the CI Index

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	2.921	1.12	41%
Real growth rate (in percent)	2.719	3.151	0.09	3%
Import coverage of reserves (in percent)	4.052	49.189	1.99	73%
Import coverage of reserves^2 (in percent)	-3.990	24.196	-0.97	-35%
Remittances (in percent)	2.022	0.011	0.00	0%
World economic growth (in percent)	13.520	3.579	0.48	18%
CI Score			2.72	100%
CI rating			Medium	

DEBT SUSTAINABILITY

A. External Debt Sustainability Analysis

10. Under the baseline scenario, all external PPG debt indicators remain below the policy relevant thresholds for the next twenty years (Figure 1). Given that fiscal sustainability concerns can arise well-beyond the standard ten-year DSA horizon, debt dynamics for each scenario are presented over the next twenty years. The PV of the PPG debt-to-GDP ratio is expected to increase gradually from 4.3 percent in 2018 to 8.8 percent in 2025 due mainly to new disbursements of loans for key infrastructure projects, and thereafter decline gradually to reach 4.6 percent by 2038. The residual in debt-creating flows is financed through excess withdrawals from the PF (Figure 3). As shown in Table 1, there is no breach for any debt indicators under the baseline scenario over the 20-year projection horizon.

11. The size of the export shock was customized to account for the structural production cycle, and that oil and gas exports will end in 2022. Petroleum production rose sharply between 2005 and 2012, only to fall between 2012 and 2018 as petroleum deposits were gradually declining. To prevent this structural production cycle from inflating the export shock and to account for the fact that petroleum production will cease in 2022, the export stress test was customized.[9]

12. Another issue affecting the export revenue shock is the existence of the PF. By law, all petroleum income is required to initially enter the PF, which is then invested in foreign assets. Hence, a negative shock to petroleum export revenue (price or quantity) automatically leads to a reduced accumulation of foreign assets and not an increase in additional external borrowing as assumed by the DSF under the stress scenario. However, to maintain evenhandedness of the DSF, the customized export stress test was kept and no adjustment was made to account for these PF rules.

13. Standardized stress tests show that a shock to the primary balance is the most extreme shock to the debt trajectory, causing a short-lived and marginal breach of the threshold for the PV of PPG debt-to-exports ratio. This reflects a highly volatile historical path of the primary balance, generating a large shock. The breaches are short-lived and marginal. The first breach occurs in 2022 (202 vs. threshold value of 180), while debt burden indicator returns to below threshold by 2023. The breach re-occurs for the second and the last time in 2024 (210 vs. threshold value of 180). The stress scenario assumes that public external debt would increase by US$183 million per year in 2019-20 to cover the financing gap. However, historically, the government has typically not resorted to debt financing, but

[9] Paragraph 66 of the *Guidance Note on the Bank-Fund Debt Sustainability Framework for Low Income Countries* (2017) specifies that "…under some exceptional circumstances, default stress parameters may not be appropriate as they may fail to capture structural breaks or idiosyncratic features in the country. In such circumstances the size of the shock and the interactions among the main variables can be customized." The export shock for Timor-Leste was customized as follows: (i) the historical standard deviation of petroleum exports was scaled by a factor equal to the ratio of future average petroleum exports (2019-2022) and past average petroleum exports (2008-2018); (ii) the standard deviation of non-oil export was not scaled; and (iii) the average of the two standard deviations was weighted by the average share of petroleum exports in 2019 and 2020 (the two years for which the shock applies). The customization implied that the initial standard deviation of 50.3 was scaled down to 10.3.

instead relied on excess withdrawals from the PF (see paragraph 2). Between 2012-2017, excess withdrawals averaged US$478 million per year, which is well above the funds needed to cover the financing gap under the stress scenario.

B. Public Debt Sustainability Analysis

14. Under the baseline scenario, the PV of total PPG debt-to-GDP ratio remains below the threshold level for the next twenty years (Figure 2). Under the 2019 budget, the PV of the debt-to-GDP remains well below the benchmark. The standardized sensitivity analysis shows that the largest shock is a commodity price shock, which leads to a sustained breach with the highest debt-to-GDP figures in 2034. However, as previously mentioned in paragraph 12, a fall in petroleum export revenue only results in a reduced accumulation of foreign assets of the PF and does not directly impact the primary fiscal balance (which is a main channel for the assumed shock in the public DSA).[10] Second, petroleum production from active fields is projected to end in 2022. Hence, as shown in the Staff Report (paragraph 8), even an oil price shock larger and more permanent than assumed in the DSF has only a marginal impact on the PF balance, and hence the ESI.[11]

C. External and Public Debt Sustainability Analysis under the Reform Scenario

15. The reform scenario does not lead to substantial changes in the results of debt sustainability analysis. The ending of excess withdrawals from the PF in 2028 leads to higher external borrowing. While the introduction of a VAT improves debt dynamics over the medium- and long-term by filling the financing gap from 2022 onwards, the net effect is an increase in the debt level over time. It is therefore relevant to conduct a debt analysis on the reform scenario as well. The result shows that the reform scenario does not fundamentally alter the debt analysis as debt levels remain contained. Importantly, under the reform scenario, the PF balance will not continue to fall but broadly stabilize (see Table 2).

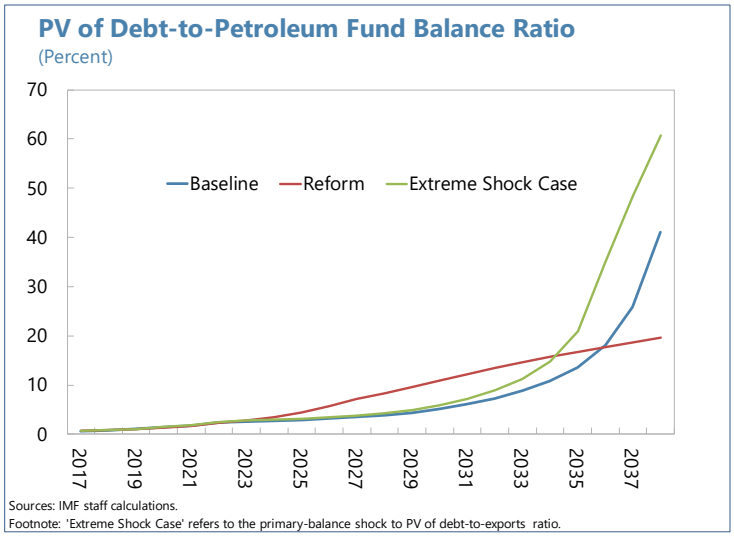
RISK RATING AND VULNERABILITIES

16. While the DSA assigns Timor-Leste a mechanical rating of "moderate" debt distress risk, judgment is applied to improve the rating to "low". The judgement reflects the country's large, liquid, and accessible net foreign assets, serving as a strong mitigating factor in Timor-Leste's ability to carry and service debt. The less ambitious frontload of public spending in the 2019 budget also improves the debt sustainability outlook.

[10] The only channel through which reduced oil prices affects budget revenue is through the ESI – which is calculated based on the total petroleum wealth, which consists of the PF balance and the projected net present value of future petroleum revenue.

[11] Moreover, the DSF assumes that the persistence of the commodity price shock is 6 years, which is well beyond the assumed end of petroleum production (i.e., 2022).

- *External debt*. Under the baseline, both solvency ratios (i.e., PV of PPG debt-to-GDP ratio and PV of PPG debt-to-exports) remain low and well below the DSA thresholds over the twenty-year projection horizon. The PV of debt is projected to climb from 6 percent of total non-oil GDP in 2017 to 24 percent in 2038. However, the PV of debt-to-exports exhibits a short-lived and marginal breach of the benchmark under extreme shock scenario (primary balance shock)—generating a mechanical debt distress rating of "moderate."
- *Public debt*. Under the baseline, both solvency ratios (i.e., PV of PPG debt-to-GDP ratio and PV of PPG debt-to-revenue) remain low and well below the DSA benchmark over the twenty-year projection horizon. As discussed above, while an extended breach of the debt-to-GDP benchmark occurs under the extreme commodity price shock scenario, the PF will insulate the budget from commodity price volatility. Also, petroleum production is projected to cease in 2022..
- *Availability of the Petroleum Fund as a buffer*. Timor-Leste has a large stock of liquid assets that should be taken into account as a buffer for the country's debt sustainability (see paragraph 2). The PF has assets at 848 percent of total non-oil GDP in 2018 and is estimated to be of a high multiple of the present and projected debt. Moreover, the PF is highly liquid with 50 percent of its assets invested in U.S. Treasuries. The recent history of annual excess withdrawals for budget financing (in accordance with the Petroleum Fund Law) indicates that these funds would be readily available to service debt if needed (see paragraph 2).[12] Indeed, the PV of debt does not exceed more than one third of the projected value of the PF over the next 20 years under the baseline scenario (see figure).[13] Under the scenario of the most extreme shock case where the PV of debt-to-exports breaches the threshold, the PV of debt-to-Petroleum Fund assets remains stable, reaching close to 60 percent only towards the end of the projection periods.



- *Overall low risk of debt distress*. With this, risk of debt distress is assessed as low, which is an improvement from the previous rating of "moderate" risk. The change reflects fuller consideration of risk mitigating elements, in line with the use of judgment in the revised LIC DSF methodology.

[12] The Petroleum Fund Law allows withdrawals in excess of the ESI provided that the Parliament approves the Government's explanation that this is made in the long-term interests of the country. The law does not impose a limit on the size of the withdrawals.

[13] The value of the PF excludes the US$650 million that are planned to be used to buy-out shares from joint venture partners in the Greater Sunrise.

17. Towards the end of and beyond the 20-year horizon, the coverage of the PF deteriorates, highlighting the need to develop a fiscal strategy to ensure long-term fiscal sustainability and safeguard the assets of the Petroleum Fund. Under current policies, the PF will deplete beyond the 20-year horizon, as the revenue base is low and capital spending is expected to increase to fill the country's much-needed infrastructure gap. Depletion of highly liquid financial assets reinforces the need for a reform strategy, which entails improving expenditure control and efficiency, mobilizing domestic revenue, and committing to protecting the wealth of the Petroleum Fund.

AUTHORITIES' VIEWS

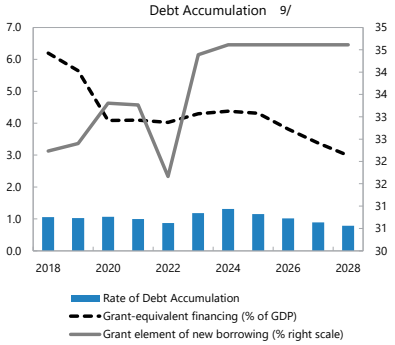
18. The authorities concurred with the overall rating of low risk of external debt distress. They welcomed the improved rating on external debt, highlighting the low level of external borrowing and the availability of the Petroleum Fund as a buffer. Furthermore, the authorities are committed to the mobilization of domestic revenues in order to protect the Petroleum Fund balance and ensure long-term debt sustainability.

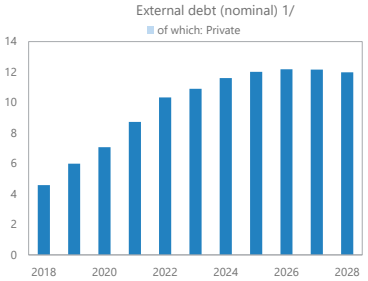
Table 4. Timor-Leste: External Debt Sustainability Framework, Baseline Scenario, 2015-2038

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2028	2038	Historical	Projections
External debt (nominal) 1/	**1.5**	**3.1**	**3.8**	**4.6**	**6.0**	**7.1**	**8.7**	**10.3**	**10.9**	**12.0**	**6.8**	**0.9**	**9.8**
of which: public and publicly guaranteed (PPG)	**1.5**	**3.1**	**3.8**	**4.6**	**6.0**	**7.1**	**8.7**	**10.3**	**10.9**	**12.0**	**6.8**	**0.9**	**9.8**
Change in external debt	0.9	1.6	0.8	0.7	1.4	1.1	1.7	1.6	0.6	-0.2	-0.6		
Identified net debt-creating flows	**-7.4**	**22.2**	**9.7**	**8.2**	**-2.4**	**0.9**	**-1.0**	**10.7**	**4.0**	**-0.9**	**-0.3**	**-9.3**	**1.3**
Non-interest current account deficit	**-6.6**	**21.6**	**10.2**	**8.9**	**-1.6**	**1.7**	**-0.5**	**10.2**	**4.9**	**-0.3**	**-0.2**	**-24.6**	**2.0**
Deficit in balance of goods and services	7.5	35.3	19.3	20.3	22.2	23.7	22.3	33.9	26.2	18.8	16.9	**-11.5**	**23.7**
Exports	34.7	12.7	19.1	16.3	11.9	14.0	14.2	8.2	9.1	11.2	14.2		
Imports	42.2	48.1	38.4	36.6	34.1	37.7	36.4	42.1	35.3	30.0	31.1		
Net current transfers (negative = inflow)	-4.0	-1.0	2.2	-2.8	-3.3	-3.6	-4.2	-4.9	-4.9	-3.5	-1.7	**-6.2**	**-4.0**
of which: official	-7.2	-7.2	-6.4	-5.8	-5.2	-4.9	-5.0	-5.2	-4.6	-2.7	-1.0		
Other current account flows (negative = net inflow)	-10.1	-12.7	-11.3	-8.6	-20.6	-18.4	-18.6	-18.8	-16.4	-15.6	-15.4	**-7.0**	**-17.7**
Net FDI (negative = inflow)	**-1.0**	**0.3**	**-0.2**	**-1.0**	**-0.6**	**-0.9**	**-1.5**	**-1.6**	**-1.4**	**-0.3**	**0.0**	**-0.7**	**-1.0**
Endogenous debt dynamics 2/	**0.2**	**0.3**	**-0.3**	**0.3**	**-0.1**	**0.1**	**1.1**	**2.1**	**0.5**	**-0.2**	**-0.2**		
Contribution from nominal interest rate	0.0	0.0	0.0	0.0	0.2	0.2	0.2	0.4	0.4	0.3	0.2		
Contribution from real GDP growth	-0.1	0.0	0.1	0.3	-0.3	-0.1	0.9	1.7	0.2	-0.5	-0.3		
Contribution from price and exchange rate changes	0.3	0.4	-0.4	...	...	...	...	...	...	...	...		
Residual 3/	**8.3**	**-20.7**	**-8.9**	**-7.5**	**3.8**	**0.2**	**2.6**	**-9.1**	**-3.4**	**0.7**	**-0.2**	**10.0**	**-0.6**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**3.4**	**4.0**	**5.0**	**5.7**	**6.9**	**8.0**	**8.2**	**8.6**	**4.6**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**17.8**	**24.4**	**41.7**	**40.4**	**48.4**	**97.6**	**90.4**	**76.4**	**32.6**		
PPG debt service-to-exports ratio	**0.0**	**0.0**	**0.5**	**0.0**	**1.3**	**1.3**	**1.7**	**4.3**	**3.9**	**4.3**	**2.8**		
PPG debt service-to-revenue ratio	**0.0**	**0.0**	**0.4**	**0.0**	**0.7**	**0.9**	**1.1**	**1.6**	**1.8**	**3.3**	**2.8**		
Gross external financing need (Million of U.S. dollars)	-233.9	551.7	280.1	248.6	-66.2	31.6	-60.0	283.2	134.1	-9.8	38.1		
Key macroeconomic assumptions													
Real GDP growth (in percent)	20.9	0.8	-4.5	-8.0	6.6	1.3	-12.3	-19.2	-2.0	4.8	4.8	**0.0**	**-0.9**
GDP deflator in US dollar terms (change in percent)	-36.5	-19.5	15.4	22.1	-5.6	4.8	11.0	19.9	13.5	6.0	6.0	**3.3**	**8.7**
Effective interest rate (percent) 4/	0.0	0.0	0.2	0.0	3.5	3.2	3.3	3.9	3.8	2.8	2.4	**0.0**	**3.0**
Growth of exports of G&S (US dollar terms, in percent)	-43.5	-70.2	65.5	-4.4	-26.5	25.3	-1.7	-44.1	23.8	16.6	11.8	**3.8**	**5.0**
Growth of imports of G&S (US dollar terms, in percent)	1.8	-7.5	-11.9	7.0	-6.2	17.3	-6.0	11.9	-6.6	7.6	14.4	**13.9**	**5.0**
Grant element of new public sector borrowing (in percent)	...	...	...	32.2	32.4	33.3	33.3	31.7	34.4	34.6	34.6	**...**	**33.7**
Government revenues (excluding grants, in percent of GDP)	26.1	29.6	24.3	23.8	22.6	20.7	21.1	21.8	19.4	14.5	14.4	**18.2**	**19.1**
Aid flows (in Million of US dollars) 5/	222.7	181.3	176.7	184.2	175.1	134.9	131.6	121.9	140.1	176.3	121.1		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	6.2	5.6	4.1	4.1	4.0	4.3	3.0	0.7	**...**	**4.3**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	88.7	85.2	81.1	80.8	81.8	80.2	79.1	72.2	**...**	**81.3**
Nominal GDP (Million of US dollars)	3,104	2,521	2,778	3,122	3,143	3,335	3,246	3,144	3,498	5,918	16,939		
Nominal dollar GDP growth	-23.3	-18.8	10.2	12.4	0.7	6.1	-2.7	-3.1	11.3	11.1	11.1	**3.4**	**7.3**
Memorandum items:													
PV of external debt 7/	...	...	3.4	4.0	5.0	5.7	6.9	8.0	8.2	8.6	4.6		
In percent of exports	...	...	17.8	24.4	41.7	40.4	48.4	97.6	90.4	76.4	32.6		
Total external debt service-to-exports ratio	0.0	0.0	0.5	0.0	1.3	1.3	1.7	4.3	3.9	4.3	2.8		
PV of PPG external debt (in Million of US dollars)			94.3	123.6	155.6	189.1	222.4	250.7	287.8	506.7	784.0		
(PVt-PVt-1)/GDPt-1 (in percent)				1.1	1.0	1.1	1.0	0.9	1.2	0.8	0.1		
Non-interest current account deficit that stabilizes debt ratio	-7.5	20.0	9.5	8.2	-3.0	0.6	-2.2	8.6	4.3	-0.1	0.4		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

9/ The grant element may be overestimated due to debt projections.

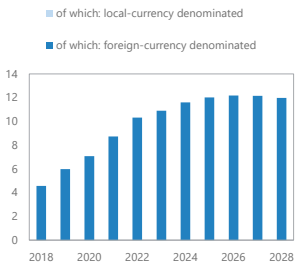
Table 5. Timor-Leste: Public Sector Debt Sustainability Framework, Baseline Scenario, 2015-2038

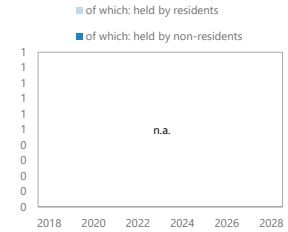
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2028	2038	Historical	Projections
Public sector debt 1/	**1.5**	**3.1**	**3.8**	**4.6**	**6.0**	**7.1**	**8.7**	**10.3**	**10.9**	**12.0**	**6.8**	**0.9**	**9.8**
of which: external debt	**1.5**	**3.1**	**3.8**	**4.6**	**6.0**	**7.1**	**8.7**	**10.3**	**10.9**	**12.0**	**6.8**	**0.9**	**9.8**
Change in public sector debt	0.9	1.6	0.8	0.7	1.4	1.1	1.7	1.6	0.6	-0.2	-0.6		
Identified debt-creating flows	**0.1**	**4.3**	**2.6**	**-1.9**	**5.9**	**8.6**	**11.1**	**13.3**	**12.4**	**10.8**	**3.6**	**-0.8**	**9.6**
Primary deficit	**-0.1**	**4.0**	**2.8**	**-1.4**	**5.8**	**8.7**	**10.7**	**12.7**	**13.1**	**11.7**	**4.2**	**-4.9**	**10.0**
Revenue and grants	33.2	36.8	30.7	29.7	27.8	24.3	24.7	25.4	23.2	17.1	14.9	**24.4**	**22.9**
of which: grants	7.2	7.2	6.4	5.8	5.2	3.6	3.6	3.6	3.7	2.6	0.6		
Primary (noninterest) expenditure	33.2	40.8	33.5	28.2	33.6	33.0	35.4	38.1	36.3	28.8	19.2	**19.5**	**33.0**
Automatic debt dynamics	**0.2**	**0.3**	**-0.3**	**-0.4**	**0.1**	**-0.2**	**0.4**	**0.6**	**-0.7**	**-0.9**	**-0.6**		
Contribution from interest rate/growth differential	-0.1	0.0	0.1	0.3	-0.2	0.0	1.1	2.3	0.4	-0.5	-0.3		
of which: contribution from average real interest rate	0.0	0.0	0.0	-0.1	0.1	0.1	0.1	0.2	0.2	0.1	0.0		
of which: contribution from real GDP growth	-0.1	0.0	0.1	0.3	-0.3	-0.1	1.0	2.1	0.2	-0.6	-0.3		
Contribution from real exchange rate depreciation	0.3	0.4	-0.4	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**0.8**	**-2.8**	**-1.8**	**1.9**	**-4.2**	**-7.6**	**-10.1**	**-13.3**	**-12.9**	**-11.4**	**-4.5**	**1.5**	**-9.4**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**3.4**	**4.0**	**5.0**	**5.7**	**6.9**	**8.0**	**8.2**	**8.6**	**4.6**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**11.1**	**13.3**	**17.8**	**23.3**	**27.7**	**31.4**	**35.5**	**50.1**	**31.0**		
Debt service-to-revenue and grants ratio 3/	**0.0**	**0.0**	**0.3**	**0.0**	**0.6**	**0.7**	**1.0**	**1.4**	**1.5**	**2.8**	**2.7**		
Gross financing need 4/	-0.1	4.0	2.9	-1.4	5.9	8.9	10.9	13.0	13.5	12.1	4.6		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	20.9	0.8	-4.5	-8.0	6.6	1.3	-12.3	-19.2	-2.0	4.8	4.8	**0.0**	**-0.9**
Average nominal interest rate on external debt (in percent)	0.0	0.0	0.2	0.0	3.5	3.2	3.3	3.9	3.8	2.8	2.4	**0.0**	**3.0**
Average real interest rate on domestic debt (in percent)	57.5	24.2	-13.2	-18.1	9.6	-1.5	-6.9	-13.3	-8.5	-3.0	-3.4	**15.4**	**-4.8**
Real exchange rate depreciation (in percent, + indicates depreciation)	60.3	25.5	-12.2	...	...	...	...	...	...	...	...	**4.2**	**...**
Inflation rate (GDP deflator, in percent)	-36.5	-19.5	15.4	22.1	-5.6	4.8	11.0	19.9	13.5	6.0	6.0	**3.3**	**8.7**
Growth of real primary spending (deflated by GDP deflator, in percent)	76.6	23.9	-21.5	-22.5	26.9	-0.3	-6.0	-13.1	-6.7	0.2	0.8	**20.2**	**-2.0**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-1.0	2.4	2.1	-2.2	4.4	7.6	9.0	11.1	12.5	11.8	4.8	**1.2**	**9.3**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government. Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 6. Timor-Leste: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2018-2028

(In percent)

	Projections 1/										
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of debt-to GDP ratio											
Baseline	4	5	6	7	8	8	9	9	9	9	9
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	4	-11	-28	-43	-65	-85	-100	-116	-131	-145	-159
B. Bound Tests											
B1. Real GDP growth	4	7	10	12	14	15	15	16	16	16	15
B2. Primary balance	4	9	13	15	17	16	16	16	15	15	14
B3. Exports	4	6	9	10	12	12	12	12	12	11	11
B4. Other flows 3/	4	8	13	14	16	16	16	15	15	14	13
B5. Depreciation	4	6	0	1	2	3	4	4	5	6	6
B6. Combination of B1-B5	4	10	6	7	9	10	10	10	11	11	11
C. Tailored Tests											
C1. Combined contingent liabilities	4	8	9	10	12	12	12	12	12	11	11
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	4	7	10	11	11	10	9	7	6	5	4
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	24	42	40	48	98	90	112	98	89	82	76
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	24	-90	-197	-301	-799	-929	-1309	-1300	-1322	-1360	-1416
B. Bound Tests											
B1. Real GDP growth	24	42	40	48	98	90	112	98	89	82	76
B2. Primary balance	24	73	93	105	**202**	178	**210**	177	154	137	125
B3. Exports	24	59	79	91	178	161	**193**	165	145	130	119
B4. Other flows 3/	24	71	91	102	**196**	174	**205**	171	147	129	116
B5. Depreciation	24	42	-1	5	17	22	36	37	40	42	43
B6. Combination of B1-B5	24	71	28	40	82	79	100	88	82	77	73
C. Tailored Tests											
C1. Combined contingent liabilities	24	70	64	73	144	130	156	134	119	108	99
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	24	80	85	89	163	120	121	89	67	51	39
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	0	1	1	2	4	4	5	5	5	4	4
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	0	1	-2	-5	-12	-17	-26	-36	-46	-54	-65
B. Bound Tests											
B1. Real GDP growth	0	1	1	2	4	4	5	5	5	4	4
B2. Primary balance	0	1	2	3	7	6	8	9	10	9	8
B3. Exports	0	2	2	3	7	6	8	8	9	8	7
B4. Other flows 3/	0	1	2	3	7	6	7	9	10	9	8
B5. Depreciation	0	1	1	1	2	2	3	3	1	1	1
B6. Combination of B1-B5	0	1	2	2	4	4	5	7	4	4	4
C. Tailored Tests											
C1. Combined contingent liabilities	0	1	2	2	5	5	6	6	5	5	5
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	0	2	2	3	7	5	6	7	7	6	5
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	0	1	1	1	2	2	2	3	3	3	3
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	0	1	-1	-3	-5	-8	-11	-19	-28	-38	-50
B. Bound Tests											
B1. Real GDP growth	0	1	2	2	3	3	4	5	5	6	6
B2. Primary balance	0	1	1	2	3	3	3	5	6	6	6
B3. Exports	0	1	1	2	2	2	3	3	4	5	5
B4. Other flows 3/	0	1	1	2	3	3	3	5	6	6	6
B5. Depreciation	0	1	1	0	1	1	2	2	0	1	1
B6. Combination of B1-B5	0	1	2	1	2	2	3	5	3	4	4
C. Tailored Tests											
C1. Combined contingent liabilities	0	1	1	2	2	2	3	3	3	4	4
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	0	1	1	2	2	2	3	3	4	4	3
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

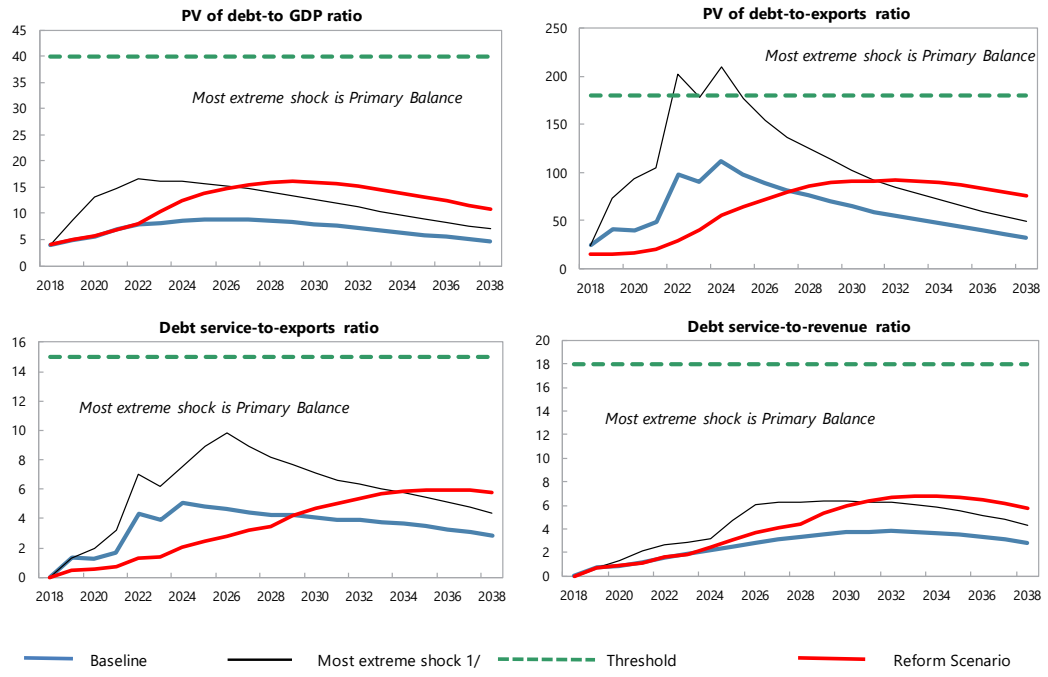
Table 7. Timor-Leste: Sensitivity Analysis for Key Indicators of Public Debt, 2018-2028

(In percent)

	Projections 1/										
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of Debt-to-GDP Ratio											
Baseline	4	5	6	7	8	8	9	9	9	9	9
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	4	-2	-11	-20	-31	-42	-53	-65	-78	-91	-105
B. Bound Tests											
B1. Real GDP growth	4	13	29	43	**59**	**67**	**74**	**80**	**84**	**87**	**90**
B2. Primary balance	4	9	13	15	17	16	16	16	15	15	14
B3. Exports	4	6	9	10	12	12	12	12	11	11	11
B4. Other flows 3/	4	8	13	14	16	16	16	15	15	14	13
B5. Depreciation	4	5	4	4	4	3	3	2	2	1	1
B6. Combination of B1-B5	4	8	8	12	13	13	13	13	13	13	13
C. Tailored Tests											
C1. Combined contingent liabilities	4	8	9	10	12	12	12	12	12	11	11
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	4	16	37	**62**	**90**	**105**	**116**	**123**	**128**	**132**	**135**
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	13	18	23	28	31	36	39	42	46	48	50
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	13	-8	-44	-82	-122	-182	-243	-310	-392	-482	-580
B. Bound Tests											
B1. Real GDP growth	13	44	106	158	210	258	299	338	384	428	469
B2. Primary balance	13	31	54	60	65	70	74	76	79	81	82
B3. Exports	13	22	36	41	45	50	53	56	59	61	62
B4. Other flows 3/	13	30	52	58	63	68	72	74	75	76	76
B5. Depreciation	13	18	18	17	15	14	12	10	8	6	3
B6. Combination of B1-B5	13	27	32	47	51	56	60	63	67	70	73
C. Tailored Tests											
C1. Combined contingent liabilities	13	30	37	42	46	51	55	58	61	63	65
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	13	64	166	272	368	454	514	557	625	689	748
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	0.0	0.6	0.7	1.0	1.4	1.5	1.8	2.1	2.4	2.6	2.8
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2018-2038 2/	0.0	0.6	0.0	-0.8	-1.6	-2.8	-4.0	-7.3	-11.6	-16.7	-22.4
B. Bound Tests											
B1. Real GDP growth	0.0	0.7	1.9	3.5	5.3	6.6	7.9	10.9	14.8	18.4	21.8
B2. Primary balance	0.0	0.6	1.1	1.8	2.3	2.4	2.6	3.8	5.1	5.2	5.3
B3. Exports	0.0	0.6	0.9	1.3	1.7	1.9	2.1	2.7	3.5	3.7	3.8
B4. Other flows 3/	0.0	0.6	1.1	1.8	2.2	2.4	2.6	3.7	4.9	5.0	5.1
B5. Depreciation	0.0	0.6	0.8	0.9	1.2	1.2	1.4	1.2	1.0	0.7	0.5
B6. Combination of B1-B5	0.0	0.6	0.8	1.1	1.6	1.8	2.0	2.4	2.7	3.0	3.2
C. Tailored Tests											
C1. Combined contingent liabilities	0.0	0.6	1.1	1.4	1.8	1.9	2.2	2.4	2.7	3.0	3.1
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	0.0	0.6	2.4	5.4	8.5	10.8	12.5	17.3	24.5	30.8	36.1
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

Figure 1. Timor-Leste: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2018-2038



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	Yes	
Natural Disasters	n.a.	n.a.
Commodity Prices 2/	No	No
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.9%	1.9%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	26	26
Avg. grace period	5	5

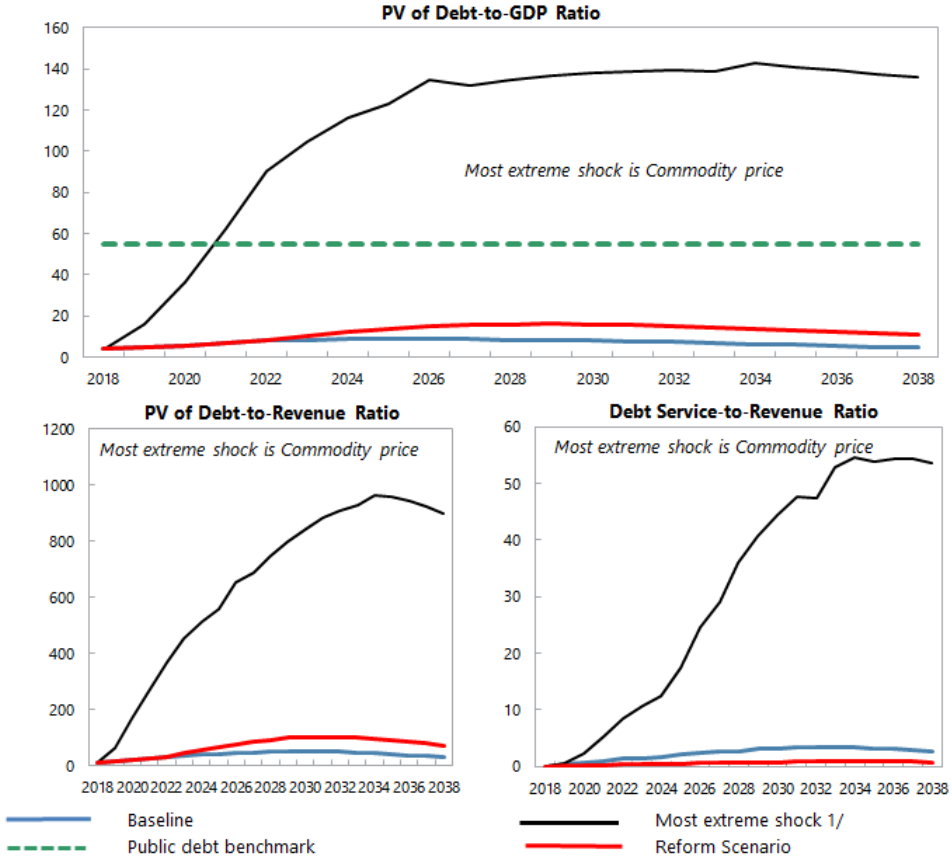
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2028. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Timor-Leste: Indicators of Public Debt under Alternative Scenarios, 2018-2038



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	100%	100%
Domestic medium and long-term	0%	0%
Domestic short-term	0%	0%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.9%	1.9%
Avg. maturity (incl. grace period)	26	26
Avg. grace period	5	5
Domestic MLT debt		
Avg. real interest rate on new borrowing	0.0%	0.0%
Avg. maturity (incl. grace period)	1	1
Avg. grace period	0	0
Domestic short-term debt		
Avg. real interest rate	0%	0.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

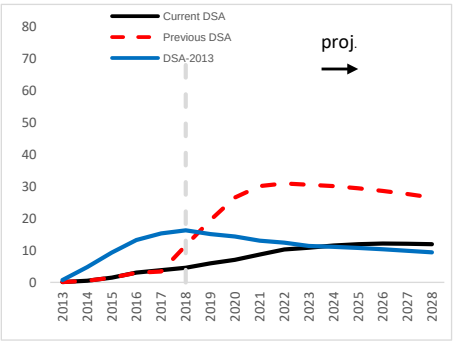
Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2028. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

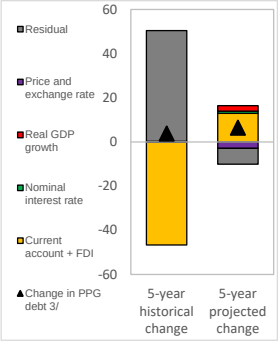
Figure 3. Timor-Leste: Drivers of Debt Dynamics – Baseline Scenario

External Debt

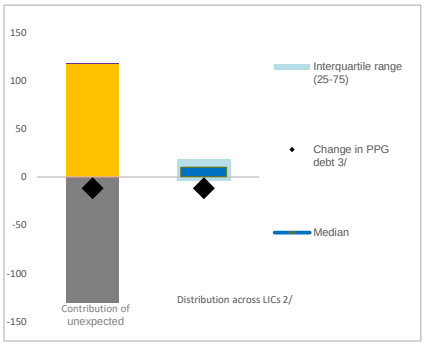
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

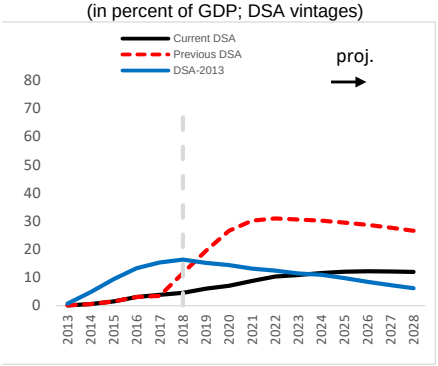


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

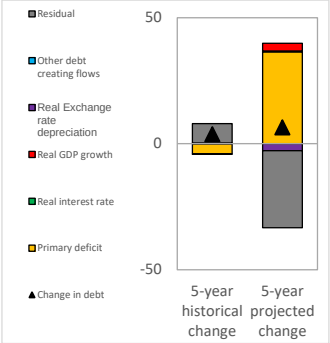


Public Debt

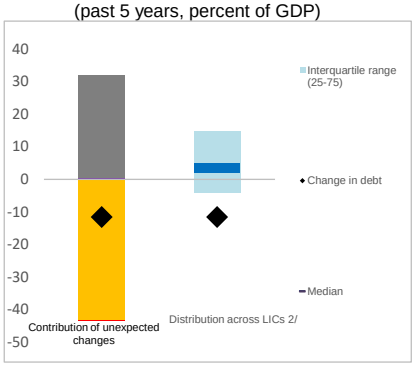
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



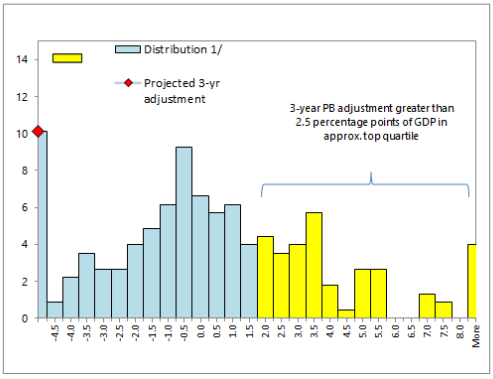
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

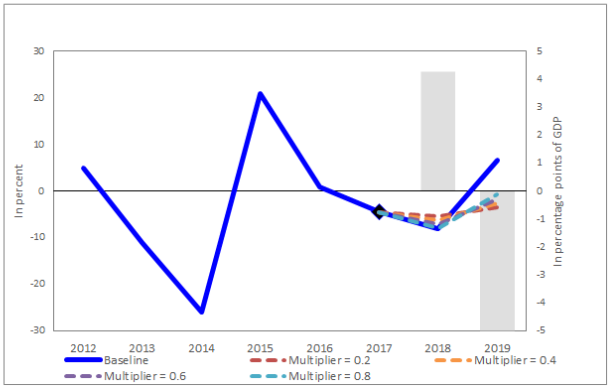
Figure 4. Timor-Leste: Realism Tools 1/

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



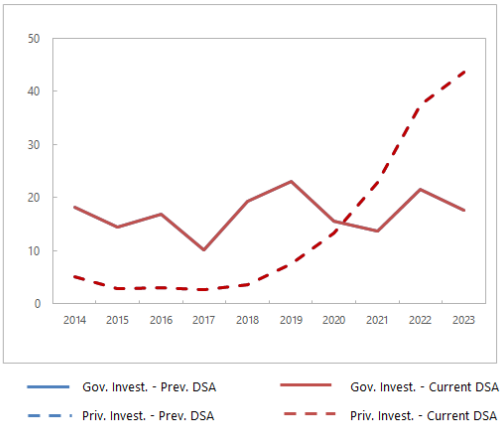
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(% of GDP)



1/ The data needed to conduct the investment growth realism tool is not available for the current DSA exercise. Efforts will be made to collect the data for the next DSA.